

This Is How Students Think About And Act Upon Personalized, Automated Feedback on Transactivity

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Abstract When people learn together, the degree to which they respond to and build on each other’s ideas matters for learning outcomes. This process, called transactivity, is a key predictor of success in collaborative settings. In educational discourse mediated by Computer-Supported Collaborative Learning applications, we can compute numeric representations of messages’ semantics (i.e., embeddings) to quantify their relatedness. This enables estimating of the degree to which participants build on each other’s ideas and thus measure transactivity. Such measurements can inform automated feedback on students’ transactivity.

One missing puzzle piece has been evaluating the effectiveness of such feedback. We present a randomized field study in which a cohort of student teachers (n=252) worked on two online discussion tasks and received one of two feedback types: *basic feedback* (BAF) with general information on transactivity and a brief performance summary, or *highly informative feedback* (HIF) with a high degree of personalization on various aspects of transactivity. We show that while HIF was perceived more positively, it did not lead to robust behavioral improvements over BAF, highlighting a striking disconnect between perceived and actual feedback effectiveness.

Keywords: Transactivity · Educational Discourse · Automated Textual Feedback · Computer-Supported Collaborative Learning · Knowledge Construction · Learning Analytics · Natural Language Processing

1 Introduction and Background

Since the advent of networked personal computing, researchers in Computer-Supported Collaborative Learning (CSCL) have been working to digitally en-

hance collaboration in education [29]. Computer-mediated discourse was one of the earliest applications: In 1985, researchers at the Gallaudet University (a university for deaf or hearing impaired students) constructed special classrooms in which they arranged networked computers in a circle, running precursors of today’s chat programs. These would then be used by the students and their instructor to conduct textual discussions [29, 4, 11]. Today, over 40 years later and with CSCL having become a *normal science* [1, 18], new methods to examine educational discussions (which are often referred to as *argumentative knowledge construction* [33]) continue to emerge. Among the enduring questions driving this research is how to foster productive discussions in the first place.

When learning in groups, learners have to take up and discuss ideas by others. This is a sign that the group considers the ideas valuable. Conversely, it is important for each member to reflect on their own ideas in light of the discussion. This way, ideas become part of the group’s co-constructed body of knowledge. One key pillar of collaborative construction of knowledge is thus transactivity, the degree to which participants respond to and build on each other’s ideas [2, 30, 33]. Reflecting and discussing ideas is one part, but for a conversation to be productive, there must also be ideas worth discussing in the first place. The other key aspect of productive discussions is thus the novelty of contributed ideas, which must be balanced with transactive behavior [31, p. 399].

There have been previous attempts to actively improve learners’ transactivity. For example, Jurkowski, Mundelsee, and Hanze [15] successfully trained secondary school students on transactive communication before letting them work on an assignment in pairs. This required them to carry out a workshop where professional teachers would train the students for two and a half days before the actual assignment. While this approach worked, it still requires extra time and a workshop outside the curriculum, or custom curriculum-integrated material on transactive communication. A different, more scalable approach would be to provide students with automatic feedback.

For feedback to be effective, it has to meet a number of requirements. [12, 35, 27]. Formative feedback in particular is known for its great potential effects, though improvement through its presence is not guaranteed [25]. This is because learners receive in-depth insights about their learning process or products rather than just a grade. For research on the effectiveness of formative feedback, this implies a necessity of at least two rounds of activity, with the feedback in between. Such an intervention can also serve as scaffolding: collaborative learning does not emerge spontaneously [17], and in CSCL, researchers frequently scaffold discussions using scripts [8]. Feedback represents another avenue to structure and support collaboration.

True formative feedback must be highly informative, actionable [6] and tailored toward learners’ goals, or as Jivet et al. [14] put it, providing students with awareness is not enough. This is why we took a different approach from, e.g., Pfannenstiel et al. [21], who provided students with a real-time widget on their transactivity during a discussion. These ideals inform the notion of *highly informative learning analytics* [10]. In that sense, *highly informative feedback*

is Learning Analytics (LA)-informed text that carries an interpretation of the indicators used and provides actionable suggestions on how to proceed [19].

For collaboration processes, this can only be provided at scale, if we can measure transactivity and newness computationally, though. Since the advent of sentence transformer language models, namely *Sentence-BERT* [22] it is possible to compute the similarity of two utterances, which can serve as a base to operationalize measures of transactivity. *Group Communication Analysis* (GCA) [9] offers a methodological framework to quantify transactivity through measure like *responsivity* (the degree to which learners engage with peers' ideas), *social impact* (the degree to which peers build on their contributions), and *newness* (the amount of information added to the existing discussion corpus). Together, these indicators make it possible to automate the detection of transactivity and newness of ideas in a discussion with a degree of precision that allows us to transform their outputs into feedback that is actionable.

In previous research, we supplied students with *highly informative feedback* on their collaboration [19] and found that learners prefer this *self-(regulation-)* and *process-focused* [cf. 3] feedback over a simpler form provided to a control group. However, that pilot study did not measure the feedback's effect on transactivity, and we identified the impact of feedback on transactivity in CSCL as a substantial gap in the literature. In the present study, we therefore examine to what extent our feedback improves students' transactivity and newness in educational discourse. Our results reveal a striking discrepancy between students' perception of the feedback and its actual effect on their behavior. We discuss the implications for the design of future feedback systems in light of this finding.

2 Research Questions and Methodology

The main goal of the present research was to investigate the impact of automated feedback on students' transactivity in a collaborative discussion assignment depending on feedback format: *highly informative feedback* (HIF), a personalized and actionable form of process feedback, as opposed to *basic feedback* (BAF). The BAF only contains information on transactivity in general and a simple summary of the students' performance. By contrast, the HIF contains information based on various indicators of transactivity and newness that are integrated into a feedback system we developed. It includes the information on transactivity from the BAF, but the summary is replaced with a performance evaluation for each indicator, as well as recommendations for actions to take in an upcoming discussion. Comparing these two formats, we address the following questions:

- RQ1a** How is *highly informative feedback* perceived by students compared to basic feedback?
- RQ1b** Does the students' perception of the feedback change across two intervention rounds?
- RQ2** Does *highly informative feedback* lead to greater improvements in transactivity and newness compared to *basic feedback*?

2.1 Experimental Design

We conducted a randomized field experiment, which we embedded in an online course in the learning management system Moodle. In educational settings, field experiments usually allow researchers to draw conclusions of higher practical value than more artificial lab experiments [23, 20]. It was structured as follows:

1. The students discussed an assigned topic in a dedicated discussion forum designed for this assignment.
2. After the assignment’s deadline, they received the feedback from our feedback generation system. They were able to review this feedback at any point in time during the course. There were two groups: the treatment group *Exp* (about half of the students in our cohort) received the HIF; the control group *Kon* received the BAF.
3. The students were instructed to review the feedback...
4. ... and to answer a questionnaire on their perception of the provided feedback,
5. ... as well as to write down what/how they want to improve in the future.
6. The course continued with a completely different kind of task.
7. After this task finished, we repeated steps 1 to 4.

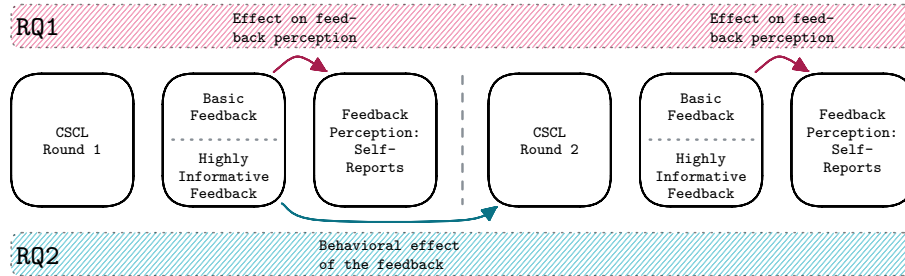


Figure 1: Timeline of the experiment

2.2 Procedure

We conducted the experiment with a cohort of undergraduate teachers in their first semester. In total, we had 283 participants in our cohort, out of which 31 had to be removed from the data set⁶, leaving us with the data of $N = 252$ participants. 110 of these were part of the treatment group. 142 were in the control group. There were 49 discussion groups with a median size of 6 students, out of which a mean of 4.95 students were included in the analysis.

⁶ Find the cleansing criteria, full assignment texts, self-report questions, feedback templates, and additional figures in the supplemental material at https://osf.io/4akjf/overview?view_only=d14e404100c743ecab7e54f79854ad79.

In a former experiment, we found that discourse indicators can depend on the micro script of an assignment [19]. Hence, to ensure comparability between the stages, both assignments followed the same scripts: first the students should watch a video and collect information about it in a Moodle discussion forum. In this phase, the students can't see what their peers post. Next, the students should discuss their collected information. Lastly, they create a joint statement as a summary of their discussion, also in the forum. They vote on the final outcome by posting that they affirm of the result.

After the deadline elapsed, the feedback was generated and a dashboard showing the feedback text and an evaluation form was unlocked in Moodle. The students had about two days to reply, before another task would begin. Due to our course design, the second stage of the experiment would not begin immediately after the first stage, though, but only after this other task.

2.3 Kinds of Collected Data

There were three main kinds of data we collected and used for our analysis:

Forum-Post-Related Data The forum posts, forum thread reply structure (i.e., parent-post markers) and reply times used to determine deadline compliance. Students who missed the deadline were excluded from the analysis, as they did not receive the correct feedback due to system constraints.

User/Group-Related Data Participants, group composition and group sizes.

Self-Reports Answers to self-report scales and free text remarks entered after reading the feedback. Our questions were based in part on the validated evaluation framework for quality learning analytics [26], as well as the derived feedback perception outcome framework by Weidlich et al. [32]. We asked the students whether ...

Agreement they felt that the feedback accurately assessed them,

Clarity if they understood what the feedback means,

Helpfulness if they found the feedback helpful for their future learning,

Motivation if the feedback motivated them,

Progress if it helped them better track their current progress,

Reflection it became easier to reflect on their learning behavior,

Regulation it became easier to adjust their learning behavior if needed,

Bias if they felt discriminated against/disadvantaged or treated unfairly,

Support and if they needed support to make this feedback useful to them.

2.4 Our Feedback Generation System

We created the feedback based on findings from our earlier work [19], where we examined the usefulness of various process data derived indicators for templating HIF about discussion quality. For the present study, we moved the implementation towards a production-ready service and reduced the indicator set to those that either directly operationalize transactivity and newness or support improving them. This leaves us with the following indicators, for which Dowell, Nixon, and Graesser [9] provide the formal definitions and the theoretical rationale:

- Participation** records how often and when a student contributed to the discussion, in relative terms. Participation is a necessary precondition for transactive interaction, as it “signifies a willingness and readiness for participants to externalize and share information and thoughts [9, p. 1010]” [5, 13].
- Responsivity** directly operationalizes transactivity by measuring how much a student responded to their fellow students’ contributions, based on the semantic similarity of their posts [13].
- Social Impact** shows how much a student’s contributions were picked up and discussed by the others. It is the flip side of responsivity and provides information about the influence their ideas had on the course of the discussion.
- Internal Cohesion** records the semantic relatedness between a student’s own contributions, reflecting the degree to which they monitor and develop their thoughts throughout the discussion [e.g., 28]. Productive participation is associated with moderate internal cohesion: neither simply reiterating the same view nor contributing unrelated points [9].
- Newness** measures the extent to which a student contributed new content or perspectives to the discussion, including their own contributions. A high value indicates that they enriched the discussion with fresh ideas [e.g., 7].

We structured the feedback templates so that each student received at least one sentence comparing their performance to the whole cohort for each indicator. If an indicator diverged from the norm enough to warrant a change, they also received a tip on how to adjust their behavior in the next discussion assignment. Finally, we included a statement based on the emergent role cluster the student fell into, derived in part from Dowell, Nixon, and Graesser [9] and described in detail in Menzel et al. [19]. Role-based feedback allows us to address patterns of behavior that go beyond individual indicators. It provides students with a more holistic picture of their overall participation profile and offers targeted suggestions for improving their transactivity. The clusters are:

- the *influential actors* (run 1: 77 → run 2: 63), who contributed early and substantively, but did not tend to engage further,
- the *drivers* (25 → 30), who maintained balanced participation throughout, combining new ideas with transactive discussion of peers’ contributions,
- the *summarizers* (86 → 75), who joined late, mostly synthesizing existing content,
- the *left-behinds* (46 → 48), who participated minimally with little impact,
- or the *chatterers* (65 → 52), whose contributions tended not to be taken up by peers, indicating off-topic or low-relevance posting.

Compared to our earlier work, we removed the *semantic uncertainty*, *density*, and *affirmation rate* indicators, and consequently the resulting *deferrers* and *cautious learners* clusters, to maintain focus on transactivity and newness.

To compute the clusters, we applied hierarchical clustering on the z-score normalized transactivity indicators in the first run, identifying clusters by inspecting the centroids. We then used centroid-based assignment in the second

run. We note that during the actual experiment, we used hierarchical clustering in both runs, a minor inconsistency that affected cluster assignments for 88 out of the 239 participants in round 2, altering the final statement in their feedback.

Our feedback structure also diverges from our pilot study [19]: there, we had one HIF template per cluster, with each template having its own conditions for texts on the indicators. Our reasoning at the time was to personalize the indicators shown based on a role-derived *learner persona*, anticipating that each persona would have different needs. Since certain indicators were excluded from the text when within an acceptable range, feedback texts could vary considerably in length under this model. For reasons of practicality and comparability, we deviated from this model in the present study. Furthermore, since the previous study did not focus primarily on transactivity, the BAF was rewritten accordingly, as described above.

3 Results

3.1 Self-Reports

Regarding the self-reports, we use (RQ1a) a direct comparison between the two treatment groups in each round and (RQ1b) a temporal comparison. To address RQ1a, we conduct a Mann-Whitney U (MWU) test between the treatment group *Exp* and the control group *Kon* for each run. We obtain an interpretable effect size r by transforming the resulting U via rank-biserial correlation. For RQ1b, we apply two complementary tests: the Wilcoxon signed-rank test on paired responses to assess within-group temporal change and MWU over each student's δ to directly contrast the magnitude of change between groups (gain score analysis). To account for the false discovery rate, we apply the Benjamini-Hochberg procedure separately to the full set of p-values yielded by each test type, across all questionnaire items and both groups jointly, at $\alpha = .05$.

When interpreted for each run individually, the analysis yielded results consistent with our expectations. Figure 2 shows that students liked the HIF much better than the BAF, which can be observed in both runs. In the Self-Report analysis, we only included students who responded in both phases, giving us 81 participants in *Exp* and 96 participants in *Kon*, totaling $N = 177$.

Throughout every category where an effect was found, the effects were in favor of the HIF. In fact, because our cohort size was large, all the significant effects that we discovered passed BH procedure over the full Mann-Whitney U derived p-value set (also figure 2). This procedure applies a very conservative threshold for null-hypothesis rejection. We are therefore confident in treating these effects as genuine. To answer RQ1a: We see small ($0.1 \leq |r| < 0.3$) to medium-sized effects ($0.3 \leq |r| < 0.5$) for all items but *agreement*, *bias*, and *motivation*. The largest effect is to be seen in the *support* item, with an $|r| = 0.46$.

To answer RQ1b, we turn to figure 3: here we can see that there are temporal, as well as treatment-induced effects present in our data. We observe the strongest effect in the *support* item: in the treatment group, where students mostly didn't

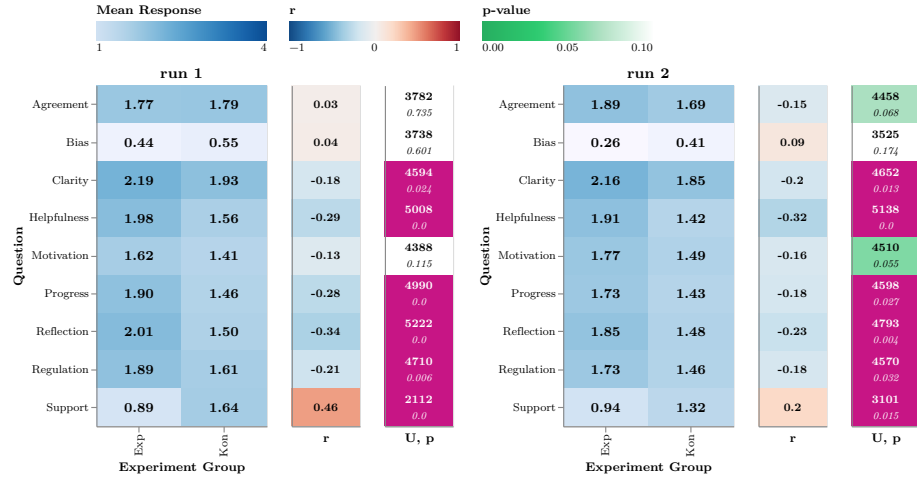


Figure 2: **Heatmap of Mean Responses by Question and Experiment Group.** For each question / group pairing, we show the mean response value and an MWU test, comparing the treatment group Exp vs. the control group Kon. Rank-biserial correlation is used as the effect size r . Magenta-highlighted cells had their H_0 rejected under the BH procedure.

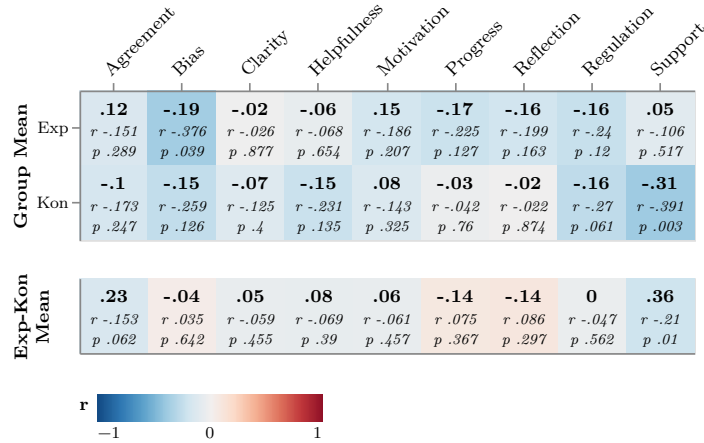


Figure 3: **Run 1 → Run 2 Mean Differences in Responses per Group.** The upper panel shows mean difference in Likert response between run 1 and run 2 ($\Delta = \text{run 2} - \text{run 1}$) per question and group, tested for significance using the Wilcoxon signed-rank test on paired responses. The lower panel shows the difference between the mean group responses (Exp - Kon), as well as test results from the MWU test on the differences per student. r is computed using rank-biserial correlation. Magenta-highlighted cells had their H_0 rejected under the BH procedure.

feel that they would need more support in the first place, that perception didn't change much in the second run. The control group, however, changed their perception significantly into the positive here, shrinking the gap between their and the treatment group's to an average of 0.38 points, down from 0.75 points, with a small effect size of $|r| = 0.21$ and $p = 0.01$ before BH correction. A similar, weaker pattern emerged for *agreement* ($p = 0.062, |r| = 0.15$). Overall it seems that students see the feedback more positively in the second round, though the effects of the HIF are mostly non-significant.

3.2 Changes in Transactivity and Newness

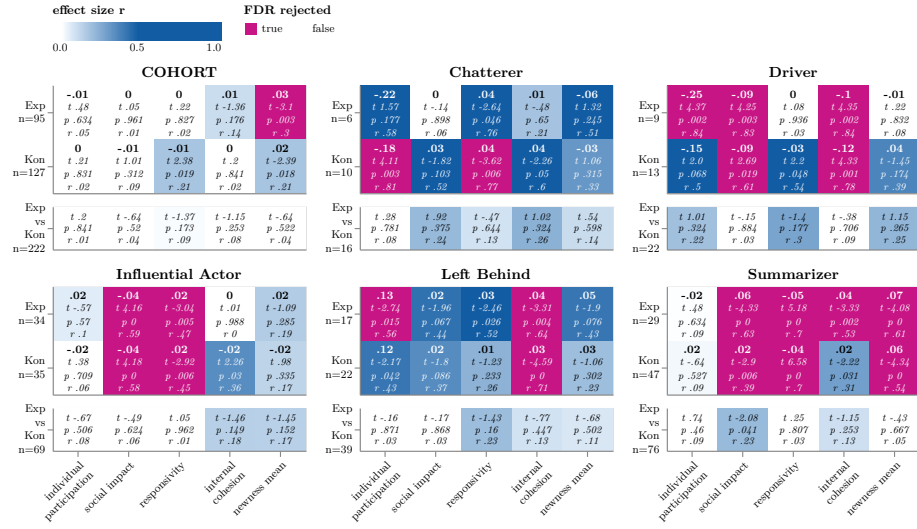


Figure 4: Heatmaps of t-test results for indicator changes between experiment runs, shown per cluster. Upper panels display mean differences, paired-sample t-statistics, p-values, and effect sizes r for treatment and control groups across five indicators. Lower panels report independent-samples t-tests comparing change magnitudes between groups. Cell shading reflects effect size (darker = higher); magenta-highlighted cells passed the BH procedure.

To address RQ2, we apply the same gain score analysis framework as in RQ1b. We analyze changes at both the cohort level and the level of individual clusters to account for effects that may cancel out in aggregate. We restrict the sample to students who completed the self-report questionnaire in the first run, treating questionnaire completion as a proxy for feedback review. This reduces the sample to $N = 222$. For within-group temporal change, we use paired-samples t-tests in place of the Wilcoxon test to reflect the continuous nature of the indicators. For the gain score analysis contrasting the magnitude of change

between *Exp* and *Kon*, we use independent-samples t-tests in place of MWU. We apply the BH procedure once over the full set of p-values for each test in the cohort-level analysis and once over the set per test for all clusters at once.

At the cohort level, we find that the *newness* increased significantly (before FDR correction) for both, *Exp* and *Kon*. Interestingly, the *responsivity* for *Kon* decreased, with a small effect size of $r = 0.21$. Strictly speaking, however, only the *newness* gains for *Exp* passed the BH procedure.

Figure 4 also shows changes in student behavior when divided up by their clusters from the first run. These paint a very different picture, showing real changes in different areas, depending on the cluster, many of which passed BH correction. The treatment with the HIF didn't make much of a difference here; additionally, some clusters were quite small, which limits the interpretability of their results. The only value that showed a significant improvement through the treatment was the *social impact* of the *summarizers*, with a small $r = 0.23$ and a $p = 0.041$, indicating that members of this cluster were motivated to start earlier in the second run. That, too, didn't pass BH correction, though.

3.3 Cluster Transitions

We used Barnard's exact test to compare the cluster transition matrices for the treatment and the control group. For each cell, the test was conducted using the contingency table 1 as an input, which resulted in figure 5:

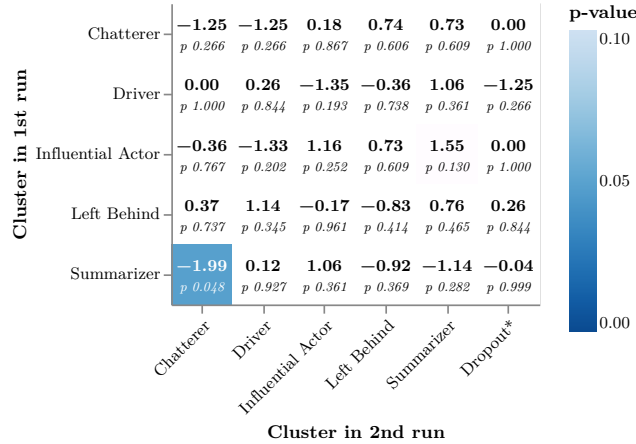


Figure 5: **Significance matrix for cluster transition probabilities between experiment runs**, comparing *Exp* and *Kon*. Each cell represents a specific cluster-to-cluster transition, tested using Barnard's exact test. A positive statistic implies relatively more transitions in *Exp* vs *Kon* and vice versa. Cell shading reflects p-values (darker = lower); magenta-highlighted cells passed the BH procedure. Cell values show the test statistic and p-value. Dropouts did not participate in the second run.

for each transition $a \rightarrow b$	Exp	Kon
transition $a \rightarrow b$	$n_{a \rightarrow b}^{Exp}$	$n_{a \rightarrow b}^{Kon}$
no transition $a \rightarrow b$	$N^{Exp} - n_{a \rightarrow b}^{Exp}$	$N^{Kon} - n_{a \rightarrow b}^{Kon}$

Table 1: Contingency table for Barnard’s exact test.

In this figure, we see that only in the *chatterers* $\rightarrow$ *chatterers* transition pair, the probability for a transition was significantly lower in the control group vs. the treatment group. That finding didn’t get its null-hypothesis rejected by the BH procedure, probably making it a false discovery. The transition analysis thus further reinforces the observation from the last section and lets us conclude that there is no robust evidence for significant changes in students’ behavior (RQ2).

4 Discussion

The study was conducted in an authentic field setting with randomized group assignments, strengthening its internal validity. The two points of measurement allowed us to examine changes attributable to formative feedback, an intervention strategy with well-documented effectiveness [25]. By combining process data with self-reports, we were able to contrast objective behavioral indicators with participants’ subjective evaluations of the feedback, allowing us to surface potential discrepancies between perception and measured behavior. These design choices enabled us to address our research questions comprehensively. The sample ($N = 177$ to $N = 252$, depending on the exclusion criteria applied) provided sufficient statistical power to detect small-to-medium effects ($r \geq 0.21$ at the lower bound, $r \geq 0.18$ at the upper bound; $1 - \beta = 0.80$, $\alpha = 0.05$) for all cohort-wide independent t-tests, and medium effects ($r \geq 0.30$ to $r \geq 0.23$, depending on group and exclusion criteria) for within-group paired sample t-tests across measurement points.

Our results show that the HIF was not able to induce a significant, robust improvement in either transactivity or newness compared to the BAF (RQ2). This was also reflected in the lack of a difference in cluster transition probability between *Exp* and *Kon*. Still, there were significant differences in student satisfaction with the feedback (RQ1a). This supports previous findings in the literature that actionable feedback is well-appreciated by learners [e.g., 6]. As a result, we could show a real discrepancy between student perceptions and the actual effect of our HIF. Furthermore, when assessed per cluster, there were strong effects between the two measurement points. These were, however, not dependent on the treatment and mostly equal between the groups (except for one significant effect that didn’t pass false discovery correction).

The temporal effects, when taken together with the convergence in students’ perceived need for additional support, are suggestive: it seems that for the outcome, the BAF with its short information text on transactivity might be suffi-

cient to change transactive behavior in forum discussion assignments. It remains unclear, however, whether external effects play a role, several of which would be plausible. One possibility is experience gains in collaborative forum tasks; another is the new assignment itself, which might have engaged students differently, despite having the same structure as the first assignment.

This highlights a limitation of our study: The absence of a second control group limits the interpretability of our results. Without a group receiving either no feedback or feedback without any information on transactivity, we cannot isolate the effects of informing students on the importance of transactivity. Furthermore, since only students who responded to both feedback forms are included in the self-report analysis, the dropout rate may introduce some distortion into the findings. We simply can't know what hasn't been reported.

That positive perceptions of feedback do not guarantee performance gains is an established finding [16, 25]. Winstone et al. [34] offer a useful framework for understanding this gap. They identify four psychological barriers to feedback uptake: *awareness*, *cognisance*, *agency*, and *volition*. Of these, the first two are unlikely explanations in our case: the HIF was written in plain language and included concrete, personalized suggestions for improvement. The more plausible barriers are agency and volition. Transactive discussion behavior is a complex social skill, which might be why dedicated training has been shown to be more effective [15]. Our feedback told students *what* they did and *how to improve*, but may not have equipped them with *strategies for how to act* differently in a live discussion. Students could revisit their feedback before the second task, and the task structure was identical. Yet deliberately changing ingrained discussion habits requires sustained effort that may exceed what students are willing or able to invest into a forum assignment.

Our results are also consistent with existing research on behavioral feedback of the kind represented by our HIF. Wisniewski, Zierer, and Hattie [35] find that of the four categories of outcome measures for feedback which they identify in their meta study, feedback on behavior was among the two categories with weaker effects, naming an effect size of Cohen's $d = 0.48$. Regarding the type of feedback given, "high-information feedback" had a large effect size, with $d = 0.99$. However, their definition of "high-information feedback" differed from our HIF, where their definition meant feedback which included *process* and *product* feedback, combined. Since our feedback is only about the process, there's no direct equivalent to our HIF in Wisniewski, Zierer, and Hattie [35].

5 Conclusion and Outlook

In our randomized field experiment, we set out to test whether *highly informative feedback* (personalized, process focused and actionable) would change students' transactive behavior compared to a *basic feedback*. Formative feedback has long been known to be effective [25], and measurable behavioral changes through feedback have been demonstrated before [35]. With this intervention, we hence had reason to expect measurable changes in student behavior. What we found

instead was a striking discrepancy between how students perceived the feedback and its actual effect on their behavior: we observed a strong preference for the HIF at both measurement points, while treatment had no significant effect.

These results suggest that educators should be cautious about relying on student perceptions as a proxy for effectiveness of HIF on transactivity and newness. One possible explanation is that these discourse properties are simply difficult to change, requiring more interventions before a clear effect emerges. However, the temporal effects we observed, albeit not in isolation, might speak against this interpretation. To disentangle these effects, future experiments could compare HIF and BAF against a no-feedback control group.

Another possible explanation for the lack of measurable behavioral change is the feedback’s degree of actionability. While the HIF contained personalized information on transactivity and newness, it did not elaborate on what exactly is being measured or why each indicator matters. Without this grounding, students may have had a target but no clear path toward it. This points to a potentially fruitful direction: greater transparency about what is measured, how it is measured, and how students are positioned relative to the relevant scales.

Based on our results, we see no reason to favor the BAF over the HIF, even in the absence of differences in behavioral outcomes. Given students’ strong and consistent preference for the HIF, we would opt for it in any long-term feedback intervention on transactivity. Student satisfaction with their educational environment matters for sustained motivation [24], and there is reason to believe that further refinements to both the underlying measurements and the feedback’s actionability, informed by continued research on HIF, can lead to improved outcomes.

Improving the quality of collaborative discussions in educational settings remains a long-standing challenge, and automated feedback is one promising avenue toward addressing it. Our findings add a nuanced piece to this endeavor: student perception and behavioral change do not always move in lockstep, and understanding this gap may be as important as closing it. As feedback systems grow more sophisticated, the question of how students receive, interpret, and act on them deserves equal attention.

6 References

AI transparency note: The authors used Google Scholar, Semantic Scholar and Claude to search for references that might become useful. All references included in this paper were screened, read and picked by the authors. We improved some phrasings in this paper using Claude and Mistral. Usually, these were iterated upon manually later in the writing process. Claude was used for proofreading and debugging code to prepare data for the analysis. Much of the Python code to render diagrams was initially drafted with Claude, but manually modified later in the process.

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